

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 What is the fundamental principle governing the propagation of light?
A) Law of gravitation B) Kirchhoff's laws C) Newton's laws of motion D) Maxwell's equations
- 2 In a cylindrical dielectric rod, light is guided primarily by which principle?
A) Total Internal Reflection (TIR) B) Diffraction C) Scattering D) Polarization
- 3 Which of the following best describes the wave model of light?
A) Light consists of particles moving in straight lines B) Light propagates in the form of electric and magnetic waves C) Light waves do not exhibit interference or diffraction D) Light waves only travel in a vacuum
- 4 According to the ray model, how does light behave when it moves from one medium to another?
A) It changes speed and direction based on the refractive indices of the media B) It always moves in a straight line C) It bends only when the second medium is a vacuum D) It gets completely absorbed at the interface
- 5 Which parameter determines the number of modes in a step-index fiber?
A) Core diameter and numerical aperture B) Wavelength of light and fiber length C) Attenuation coefficient and refractive index D) Fiber bending radius and light intensity
- 6 Which of the following is NOT a major type of dispersion in optical fibers?
A) Modal dispersion B) Chromatic dispersion C) Rayleigh dispersion D) Polarization mode dispersion
- 7 Which fabrication method is commonly used for manufacturing optical fibers?
A) Vapor Axial Deposition (VAD) B) Lithography C) Plasma etching D) Chemical vapor etching
- 8 What does an Optical Time-Domain Reflectometer (OTDR) primarily measure?
A) The temperature of an optical fiber B) The amount of light passing through the fiber C) Fiber loss, breaks, and defects along the fiber length D) The chromatic dispersion of an optical signal

- 9 Which of the following is a major factor that affects fiber splicing loss?
 A) Fiber core diameter mismatch B) Cladding material composition C) Input laser power D) Type of optical detector used
- 10 In a laser diode, population inversion is achieved by
 A) Spontaneous emission B) Electrical pumping C) Absorption of photons D) Thermionic emission
- 11 The quantum efficiency of a photodetector is defined as
 A) The ratio of output current to input optical power B) The ratio of electron-hole pairs generated to incident photons C) The ratio of absorbed photons to incident photons D) The ratio of input power to output power
- 12 The primary source of noise in an optical receiver using an APD is
 A) Shot noise B) Dark current noise C) Thermal noise D) Multiplication noise

Q2. Solve the following.

- A) Explain how the speed of light changes when it propagates through different optical media. 6
 B) Explain the concept of phase velocity and group velocity in wave propagation. 6

Q3. Solve the following.

- A) A multimode step-index fiber has a core diameter of 60 μm , a numerical aperture (NA) of 0.20, and operates at a wavelength of 850 nm. Calculate the number of guided modes supported by the fiber (approximate using $M \approx V^2/2$). 6
 B) Explain various types of dispersion in optical fibers and their effects on signal transmission. 6

Q4. Solve Any Two of the following.

- A) Describe the process of optical fiber fabrication using the Modified Chemical Vapor Deposition (MCVD) method. 6
 B) A GaAs LED emits at $\lambda = 870$ nm. It is driven with a current of 30 mA and has an internal quantum efficiency of 60%.
 (a) Find the optical power generated internally.
 (b) If only 70% of internally generated power is emitted from the device, find the emitted optical power.
 Use: $q = 1.6 \times 10^{-19}$ C, $h = 6.63 \times 10^{-34}$ Js, $c = 3 \times 10^8$ m/s. 6
 C) Define responsivity in a photodetector. Derive an expression for the responsivity of a photodiode in terms of quantum efficiency and wavelength. 6

Q5. Solve Any Two of the following.

- A) Define a Raman amplifier and explain its working principle with the help of a schematic diagram. 6
 B) Explain the working principle of Wavelength Division Multiplexing (WDM). 6
 C) Explain the concept of coupled mode theory and its application in directional coupler. 6

Q6. Solve Any Two of the following.

- A) Explain the concept of soliton-based communication. How do solitons maintain their shape while propagating through an optical fiber? 6
 B) Describe the role of self-phase modulation (SPM) and Group Velocity Dispersion (GVD) in pulse broadening and compression. 6
 C) Compare self-phase modulation (SPM) and cross-phase modulation (XPM). 6